

# Preference Without Benefit: Depth-Enhancement Shaders for Cerebral Vasculature in Passthrough Mixed Reality

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**Abstract.** Judging the depth of cerebral vessels is central to neurosurgical planning, and depth-enhancement shaders such as aerial perspective (fog) and pseudo-chromadepth measurably improve such judgements on 2D displays. Whether they help in video see-through (passthrough) Mixed Reality (MR) — where virtual anatomy competes with a real background, but binocular disparity and motion parallax are also available — is open. We report a within-subjects study ( $n = 12$ ) on a standalone Meta Quest 3. Participants viewed a cerebral-vasculature mesh embedded in a brain shell, opened by a virtual probe that cut an aperture into the shell, and performed a two-stage task: identify a designated marker among three candidates, then place a marker at its remembered location once it was hidden. We first establish that the task is depth-limited: placement error along the view axis was approximately  $2.4\times$  larger than per-axis lateral error, in all 12 participants without exception (Wilcoxon  $W = 0$ ,  $p = 0.00049$ , the minimum attainable at  $n = 12$ ). Against that sensitive task, no shader produced any detectable effect on any performance outcome. Subjectively, however, participants diverged sharply: 7 of 10 single-choice respondents named pseudo-chromadepth the most helpful shader (binomial  $p = 0.020$ , exploratory), none chose fog, and Phong was rated worst on all six workload subscales. Users preferred a cue that did not measurably help them. We argue this dissociation arises because viewing was unconstrained: with free head movement over world-locked anatomy, motion parallax supplies the depth information the shaders would otherwise carry, leaving no headroom for a shader to act in. The study is exploratory and not pre-registered.

**Keywords:** Mixed reality · Passthrough video see-through · Depth perception · Cerebral angiography · Subjective preference · Neurosurgical planning.

## 1 Introduction

Neurovascular interventions — including the planning and clipping of aneurysms, the resection of arteriovenous malformations (AVMs), and the management of arteriovenous fistulae (AVFs) — depend on a precise, three-dimensional understanding of how target vessels are situated within the surrounding brain [12]. A

central difficulty is judging the *depth* of a vessel of interest: it is typically embedded within, and partially occluded by, other anatomy, so its position relative to neighbouring vessels is hard to recover from the two-dimensional slices or flat desktop renderings used in conventional review, which provide weak and often ambiguous depth information [12].

Video see-through (passthrough) Mixed Reality (MR) is an appealing medium for this task: a head-mounted stereoscopic display supplies binocular disparity and head-coupled motion parallax — two of the strongest depth cues — while registering the vasculature to the clinician’s real workspace keeps the physical environment as context. On 2D displays, depth-enhancement shaders such as aerial perspective (fog) and pseudo-chromadepth substantially improve depth judgements of vascular data [10, 16]. In immersive VR, by contrast, stereo and parallax are so strong that shader differences largely wash out [8, 19]. Passthrough MR has been argued to sit between these regimes: the virtual anatomy competes with a real, textured background, and the passthrough imagery is geometrically distorted, reduced in resolution, and latent — the depth and occlusion problems long recognised in medical AR [15, 1]. On that reasoning, shaders should have more room to help here than in VR.

We tested that expectation and did not find it. This paper reports a within-subjects study ( $n = 12$ ) on a standalone Meta Quest 3 comparing three shaders — Phong (reference), fog, and pseudo-chromadepth — on an absolute in-scene placement task. Our contributions are:

- A *standalone* passthrough-MR system that renders a cerebral-vasculature mesh inside a brain shell, world-locked to the workspace, with a probe-driven aperture that reveals the vasculature in situ (Section 3).
- A demonstration that the placement task is genuinely *depth-limited*: view-axis error is approximately  $2.4\times$  per-axis lateral error, in every participant (Section 5.1). This matters because a null on an insensitive task is uninformative; ours is not.
- The central finding — a *dissociation*. No shader affected any performance outcome, yet participants strongly preferred pseudo-chromadepth and rated Phong as most demanding on every workload subscale (Sections 5.2–5.3).
- An account of why: with unconstrained viewing of world-locked anatomy, motion parallax already supplies the depth information, leaving no headroom for a shader (Section 6).

## 2 Related Work

**Depth cues for vascular visualisation.** For rendered vasculature, local (Phong) surface shading is the conventional illumination baseline; on a surface mesh its normals are well defined, so it is a fair reference rather than a source of gradient noise. Added depth cues improve on it. Aerial perspective (fog) encodes distance as desaturation/attenuation, and pseudo-chromadepth — a depth-to-colour (red  $\rightarrow$  blue) mapping introduced for angiography [16] — is consistently

rated among the strongest static cues for vascular data [10]; it has been adapted specifically to vascular *surface* models [3], the setting we adopt. These results motivate our three-condition set. Depth-cue comparisons are well developed for surface renderings on conventional and VR displays [9], but have not been transferred to registered, passthrough-MR viewing of cerebral vasculature.

**Depth perception in immersive VR and the ceiling effect.** Much of that evidence comes from 2D displays. In stereoscopic VR with free head movement, disparity and motion parallax are themselves powerful cues, and several studies report that shader choice then matters little: depth judgements of vascular models were more precise in VR than on desktop yet barely distinguished shading methods [8], interactive cues had little impact on relative-depth perception for angiographic volumes in VR [19], and, once camera motion is permitted, depth perception depends more on scene content than on the chosen technique [14]. Our result is best read as an extension of this ceiling effect rather than a contradiction of it (Section 6). Methodologically, 3D-pointing studies show that error along the view (depth) axis behaves differently from lateral error and is the axis on which depth cues act [18, 2], motivating an error decomposition rather than a single 3D distance — and, in our case, supplying the validity check of Section 5.1.

**Depth perception in passthrough augmented and mixed reality.** Passthrough viewing changes the depth problem rather than easing it. The depth of virtual content overlaid on a real scene is systematically misperceived, modulated by display type, shadows, and relative position [1] and by object shape, colour, and luminance [4]; without correct occlusion, virtual structures appear to float in front of real ones, a long-standing, only partially solved problem [15, 17]. These effects are acute for medical overlays, and depth-enhancing vascular visualisation has been pursued specifically for neurovascular AR [11, 13].

**Preference versus performance.** That users *prefer* a visualisation is not evidence that it helps them. Prior work on vascular depth cues has reported subjective ratings alongside performance and found pseudo-chromadepth rated highly [10], and the general possibility that preference and performance dissociate is well known in visualisation evaluation. Our study is, to our knowledge, the first to report such a dissociation for depth-enhancement shaders in passthrough MR, and — because we first establish that the task is depth-sensitive — to be able to attribute the null to the viewing regime rather than to an insensitive measure.

### 3 System

#### 3.1 Anatomical Model

The stimulus is a surface mesh of the prominent cerebral vasculature, segmented from a single patient scan and registered to the ICBM-152 template brain [6],

which supplies the outer brain shell. The model is therefore a *composite*: one patient’s vessels in a population-average brain, not patient-specific anatomy. This is adequate for a perceptual study, where the requirement is plausible and consistent geometry rather than clinical fidelity, but it bounds the clinical reading of our results (Section 6.1). Rendering an opaque surface mesh rather than performing direct volume rendering is what makes the system run untethered: ray-marched DVR of a brain volume is bandwidth- and fragment-shader-bound and has been sustained only on PC-tethered or externally accelerated systems [5]. A mesh also removes the transfer function as a confound — with no opacity ramp, occlusion is simply correct surface occlusion, held constant across conditions. **[TODO: state the scan modality (MRA/CTA/DSA), the segmentation and surface-reconstruction pipeline, and the registration method. Not recoverable from the logs.]**

### 3.2 Depth-Enhancement Shaders

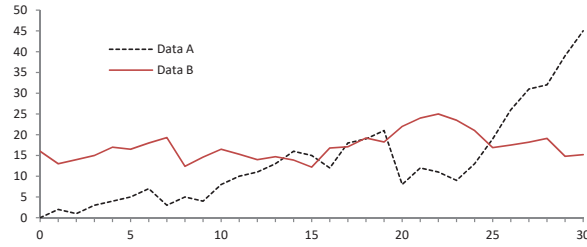
The three conditions share a single base material — surface colour and Phong lighting response — and differ only in the added distance encoding: *Phong*, standard local surface shading, the reference; *fog*, an explicit parameterised depth→desaturation applied post-shading and disabled in the other conditions; and *pseudo-chromadepth*, a depth-dependent red→blue colour mapping [16, 3]. Because the base material and occlusion are identical across conditions, each shader changes only how distance is encoded, not the underlying surface. In passthrough, fog blends toward a fixed reference colour rather than the variable real background, so it is not confounded with scene content. **[TODO: report the fog and chromadepth parameters (near/far planes, colour endpoints) in full.]**

### 3.3 Probe and Aperture

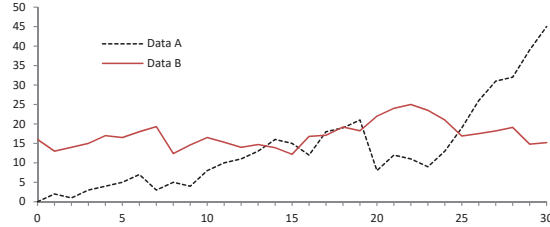
The vasculature is enclosed by the opaque brain shell and is not visible from outside it. Bringing a hand-held virtual probe near the shell cuts a small aperture into it, which remains open, revealing the vessels within; the probe may be advanced inside the shell. This deliberately mimics the constrained view of a neurosurgical approach: a small opening, a target beneath it, and an instrument reaching through. The tip of the probe is the interaction point for both stages of the task (Section 4.1).

### 3.4 Passthrough Compositing and Registration

The system is built in Unreal Engine **[TODO: confirm version; the previous draft said 5.7.3]**. The model is drawn over the Quest 3 passthrough feed using Meta’s Passthrough API and world-locked to the participant’s desk with a Meta Spatial Anchor, so it remains static as the participant moves. We stress that this anchor is a *software* world-lock, not a physical object: it does not supply



**Fig. 1.** The brain shell and enclosed vasculature composited over Quest 3 passthrough and world-locked with a Meta Spatial Anchor, with the probe-cut aperture revealing the vessels (in-headset capture). [TODO: replace placeholder.]



**Fig. 2.** The three depth-enhancement shaders applied to the *same* base material and geometry: (a) Phong (reference), (b) fog, (c) pseudo-chromadepth. [TODO: replace placeholder.]

a real surface at a known distance. What it does supply is veridical *motion parallax*, which proves central to interpreting our result (Section 6). The model was re-anchored per session, so its pose differs between participants; all reported geometry is computed within-participant.

## 4 Study Design

### 4.1 Task

On each trial, three small white markers were displayed on the vessel surface at differing depths. Participants were instructed: “*Select the marker in the middle of the three dots.*” They selected one with the probe tip and confirmed with a button press, at which point **the markers were hidden** and they placed a marker at the remembered location of the one they had selected. The placement stage is therefore *open-loop*: participants could not see the target while placing, and so could not converge on it by visual alignment. This is what makes placement a depth judgement rather than a visuomotor tracking task, and it is confirmed empirically in Section 5.1. Ground truth is the position of the designated marker; error is decomposed, using the headset pose at placement, into a component along the view axis (*depth*) and an orthogonal *lateral* component [18, 2].

## 4.2 Conditions and Design

The single manipulated factor was the shader, applied within subjects. Each participant completed 18 trials: six target locations, each presented once under each of the three shaders. Shader is thus crossed with target by construction and cannot be confounded with target identity. Presentation order was shuffled per participant (12/12 distinct orders; no order effect detectable, Spearman  $\rho = +0.09$ ,  $p = 0.23$ ). All participants saw the same target locations, re-anchored per session.

## 4.3 Excluded Trials

One target block (trials 15–17, three trials per participant) is excluded from all analyses, leaving  $12 \times 15 = 180$  trials in a balanced 5/5/5 design. The ground-truth point for that block is not a plausible target: it lies 8.14 units from the nearest other marker in the stimulus, against a median nearest-neighbour marker spacing of 2.68 units — roughly  $3\times$  off-cloud. Because the model is rigidly re-anchored per session and distances are invariant under rigid transforms, these two figures are *identical to two decimal places for all twelve participants*, which identifies the cause as a single authoring fault in the stimulus rather than a per-session error. The block accordingly shows 0/36 selection accuracy across every participant — a deterministic outcome, not a behavioural one. The exclusion is on apparatus grounds and is independent of the shader comparison; the block was uniformly faulty across all three conditions.

## 4.4 Participants and Procedure

Twelve participants were drawn from a non-clinical population, on the rationale that the core depth judgement is a basic visual-system function and that prior work found similar novice/expert cue rankings for it [10]. Five wore glasses. Most were VR novices (seven had used a headset only once or twice; two weekly, one monthly, two yearly). Six reported no familiarity with brain anatomy, three some, one detailed. Sessions took approximately 25 minutes. **[TODO: training block details; ethics approval reference.]**

## 4.5 Analysis

**This study was not pre-registered.** The emphasis on view-axis depth error as the outcome of interest was fixed in advance in consultation with the supervisor, and follows the 3D-pointing literature in which depth is the axis on which cues act [18, 2]; but there is no timestamped public registration, and we therefore report all outcomes as *exploratory* and privilege none of them inferentially. All outcomes examined are reported below, including those that are null. Because  $n = 12$  and the per-participant distributions are not assumed normal, we use Friedman omnibus tests with Wilcoxon signed-rank post-hocs under Holm

**Table 1.** Performance outcomes (per-participant means,  $n = 12$ ). Errors in scene units [TODO: confirm these are cm]; times in seconds. No outcome is significant.

| Outcome               | Phong  | Fog    | Chromadepth | Friedman                   |
|-----------------------|--------|--------|-------------|----------------------------|
| Selection accuracy    | 0.867  | 0.850  | 0.900       | $\chi^2 = 1.53, p = 0.466$ |
| Depth error, unsigned | 0.315  | 0.368  | 0.322       | $\chi^2 = 0.50, p = 0.779$ |
| Lateral error         | 0.204  | 0.228  | 0.261       | $\chi^2 = 2.00, p = 0.368$ |
| Depth error, signed   | -0.069 | -0.167 | -0.036      | $\chi^2 = 1.17, p = 0.558$ |
| Time to place (s)     | 6.85   | 8.98   | 7.71        | $\chi^2 = 1.17, p = 0.558$ |
| Time to select (s)    | 21.57  | 20.13  | 26.57       | $\chi^2 = 2.67, p = 0.264$ |

correction. We note throughout that with  $n = 12$  the smallest two-sided  $p$  attainable by any distribution-free test is  $2/2^{12} = 0.00049$ ; any smaller value would be an artefact of a parametric assumption rather than something the data can support.

## 5 Results

### 5.1 The task is depth-limited

A null result is only informative if the task could have detected an effect. We therefore first ask whether placement error is dominated by depth at all, or is merely isotropic hand noise — which, having no preferred axis, could not be improved by any depth cue.

We decomposed each error vector in a view-aligned orthonormal frame (one axis along the view direction, two perpendicular) and compared RMS error along each. This makes no distributional assumption. Depth RMS error was 0.429 against a per-axis lateral RMS of 0.193: a ratio of 2.37 (median 2.41). The effect is unanimous — *all twelve* participants show depth error exceeding lateral error (Wilcoxon signed-rank  $W = 0, p = 0.00049$ , which is exactly the floor at  $n = 12$ ). The evidence here is the unanimity and the magnitude; the  $p$ -value merely confirms that twelve-for-twelve is not a coin flip.

Depth is thus roughly  $2.4\times$  harder than the lateral axes in this task, exactly as the pointing literature predicts [18, 2]. The task has a large, depth-specific error component available for a shader to reduce.

### 5.2 No shader affected performance

No shader reduced it. Table 1 reports every performance outcome examined; all are null, and no pairwise contrast survives Holm correction.

Signed depth error was slightly negative in all conditions (grand mean  $-0.088$ ), the direction expected for the depth compression characteristic of head-mounted displays, but did not differ from zero ( $t = -1.45, p = 0.174$ ).

**Table 2.** Workload, 7-point TLX-style instrument (per-participant means,  $n = 12$ ; higher is worse). Phong is rated worst on every subscale; none is significant.

| Subscale        | Phong | Fog  | Chromadepth | Friedman                   |
|-----------------|-------|------|-------------|----------------------------|
| Mental Demand   | 4.33  | 3.50 | 3.42        | $\chi^2 = 1.37, p = 0.505$ |
| Physical Demand | 3.75  | 3.33 | 3.58        | $\chi^2 = 2.70, p = 0.260$ |
| Temporal Demand | 2.25  | 1.92 | 1.50        | $\chi^2 = 3.56, p = 0.168$ |
| Performance     | 3.83  | 3.42 | 3.50        | $\chi^2 = 1.15, p = 0.562$ |
| Effort          | 4.17  | 3.50 | 3.42        | $\chi^2 = 5.52, p = 0.063$ |
| Frustration     | 3.83  | 2.75 | 2.75        | $\chi^2 = 3.71, p = 0.156$ |

### 5.3 But participants strongly preferred chromadepth

Subjective responses diverge sharply from performance. Asked which shader they found most helpful, 7 of the 10 participants who named a single condition chose pseudo-chromadepth and 3 chose Phong; *none* chose fog (binomial against 1/3:  $p = 0.020$ ). Counting the two further participants who named several conditions, chromadepth was named by 9 of 12.

**This item is damaged and we report it with that caveat.** The response options were edited partway through data collection, so one condition appears under two labels (merged before tallying), and the item permitted multiple selections rather than forcing a choice. It is also one significant result among approximately thirteen tests in an unregistered study, and would not survive Bonferroni correction across that set ( $0.05/13 = 0.0038$ ). We therefore treat it as exploratory and suggestive, not confirmatory.

Workload responses corroborate the direction. On a seven-point TLX-style instrument [7] (higher is worse on every subscale; the Performance subscale correlates positively with the other five,  $r = +0.39$  to  $+0.79$ , confirming it is not reverse-coded here), Phong scored worst on *all six* subscales (Table 2). No subscale reaches significance — Effort is closest ( $p = 0.063$ ) — and the subscales are correlated, so this is corroborating rather than independent evidence. But the direction is unanimous.

## 6 Discussion

**A dissociation, and why it arises.** Our participants preferred a cue that did not measurably help them. The task was demonstrably depth-limited ( $2.4\times$ , unanimous), so the null is not an artefact of an insensitive measure; and the preference is consistent across two instruments that point the same way. The natural explanation is that *viewing was unconstrained*. Participants moved freely and took as long as they wished: head-position range was a median  $2.08\times$  the full extent of the marker cloud, and selection times reached 235s on some trials. Against world-locked anatomy, free head movement makes motion parallax veridical and abundant — and parallax is precisely a depth cue. The shaders



were therefore competing against a cue that had already resolved the depth, and had no headroom in which to act.

This makes our result an *extension* of the VR ceiling effect [8, 19, 14] rather than a contradiction of it. The expectation that passthrough MR would restore a role for shaders — because the imagery is degraded and the background cluttered — was not borne out: when parallax is freely available, passthrough behaves like VR in this respect. It is worth being explicit that this is not the result we expected, and that our own framing (Section 1) predicted otherwise.

Why, then, the preference? A plausible reading is that chromadepth reduces the *effort* of a judgement that participants could make regardless — consistent with Effort being the subscale closest to significance and with Phong scoring worst on all six. A cue can make a task feel easier without making it more accurate. For system designers this is not nothing: a visualisation that lowers perceived workload at no cost to accuracy may still be worth adopting. But it should not be sold as an accuracy aid on this evidence.

## 6.1 Limitations

Several limitations bound these conclusions, and the first is the most important.

*Viewing was unconstrained.* We did not restrict head movement or impose a response deadline. Because free viewing supplies parallax, our design cannot separate “shaders do not help in passthrough MR” from “shaders do not help *when parallax is available*”. We believe the latter, and the head-movement data support it, but the study cannot decide between them. This is the single change we would make first, and it is arguably the clinically relevant one: a surgeon at a fixed vantage over a craniotomy cannot orbit the anatomy, so the parallax-limited regime — not ours — is the one that matters most.

*The selection stage is not a depth measure.* The instruction “select the marker in the middle of the three dots” does not specify *middle in depth*, and the app’s ground truth defines “middle” in the model’s coordinate frame rather than along the view axis. The two come apart: the designated marker is the true view-axis middle on only 65% of trials, and on 60% of trials a second marker also satisfies the model-frame criterion, so the item often has more than one defensible answer. Selection accuracy is accordingly reported descriptively only, and should not be read as a depth-ordering judgement. The placement stage, which is our actual outcome, is unaffected by this: it requires only that the participant aims at the marker they chose.

*The sample is small and the study exploratory.* With  $n = 12$  and no pre-registration, we can claim only that we *could not detect* a difference, never that none exists; the confidence intervals readily admit effects that would matter clinically. The preference result in particular is one significant test among many and needs independent confirmation.

*The scene is fully virtual.* Both the brain shell and the vasculature are rendered; no physical object was present. The world-lock supplies parallax but not a real surface at a known distance, so our results do not speak to the case where virtual anatomy is registered to a physical phantom or patient.

*The model is a composite.* One patient’s vessels registered to a template brain is adequate for perceptual work but is not patient-specific anatomy, and the study speaks to depth judgement of a vascular surface in situ rather than of vessels seen through translucent parenchyma.

## 7 Conclusion

We built a standalone passthrough-MR system that renders cerebral vasculature inside a probe-openable brain shell on a Meta Quest 3, and used it to compare Phong, fog, and pseudo-chromadepth on an absolute depth-placement task ( $n = 12$ ). The task is demonstrably depth-limited — view-axis error is roughly  $2.4\times$  per-axis lateral error, in every participant — yet no shader improved any performance outcome. Participants nonetheless preferred pseudo-chromadepth and rated the Phong reference as the most demanding condition on every workload subscale. We attribute the null to unconstrained viewing: against world-locked anatomy, free head movement supplies the parallax that leaves shaders no headroom.

The clear next step follows directly from that account: constrain the viewpoint, and test whether shaders recover a measurable benefit when parallax is unavailable. Registering the anatomy to a physical phantom, rather than presenting it fully virtually, is the natural companion change.

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